Fluid Mechanics Notes

Marcus Koschmieder Krogsgaard

February 27, 2026

1 Introduction and Flow Kinematics

1.1 Streamlines, pathlines and streaklines

A **streamline** is a curve that is locally tangent to the velocity field at every point. This requirement can be expressed as:

$$dx/u = dy/v = dz/w \tag{1.1.0.1}$$

Streamlines can be found by using this relation to establish one or more differential equations.

A **pathline** is the curve obtained by following the trajectory of a single particle over time. To find the pathlines, use this relationship between the derivatives of the position of the particle and the velocity field

$$\frac{d}{dt}\mathbf{x}_{\text{particle}} = \mathbf{u} \tag{1.1.0.2}$$

And integrate with the initial conditions $\mathbf{x}(t_0) = (x_0, y_0, z_0)$ to get

$$\mathbf{x}_{\text{particle}}(t) = \mathbf{x}_{\text{particle}}(t, x_0, y_0, z_0, t_0) \tag{1.1.0.3}$$

A **streakline** is the curve obtained by connecting all fluid particles that will ever pass through a given point in space. The equations for streaklines are obtained in a similar way to those for pathlines

$$\mathbf{x}_{\text{streakline}}(t) = \mathbf{x}_{\text{streakline}}(t, x_0, y_0, z_0, t_0) \tag{1.1.0.4}$$

The important distinction between eqs. (1.1.0.3) and (1.1.0.4) is that t_0 is kept fixed in (1.1.0.3), since we are only considering one particle (the one that is released at t_0). In (1.1.0.4), we vary t_0 to obtain the trajectories of all particles that will ever pass through (x_0, y_0, z_0).

1.2 Lagrangian and Eulerian descriptions

- **Lagrangian:** follows the trajectory of a *specific particle* over time (particle description of motion)

- **Eulerian:** describes the motion of an arbitrary particle *at a particular location and specific instant of time* (field description of motion)

Motivation for the Eulerian description:

The Eulerian approach solves the question: *Given a velocity field* $\mathbf{V}(x, y, z, t)$*; find the acceleration* $\mathbf{a}_p$ *of a fluid particle at* (x, y, z, t)*.*

We start by noting that the instantaneous velocity of the particle is given by the velocity field at (x, y, z, t)

$$\mathbf{u}_p(t) = \mathbf{V}(x, y, z, t) \tag{1.2.0.1}$$

At $t + dt$ the particle will have a new instantaneous velocity

$$\mathbf{u}_p(t + dt) = \mathbf{V}(x + dx, y + dy, z + dz, t + dt) \tag{1.2.0.2}$$

The infinitesimal change in the particle's velocity is given by the chain rule

$$d\mathbf{u}_p = \frac{\partial \mathbf{V}}{\partial x}dx + \frac{\partial \mathbf{V}}{\partial y}dy + \frac{\partial \mathbf{V}}{\partial z}dz + \frac{\partial \mathbf{V}}{\partial t}dt \tag{1.2.0.3}$$

So the total acceleration of the particle is

$$\mathbf{a}_p = \frac{d\mathbf{V}}{dt} \tag{1.2.0.4}$$

$$= \frac{\partial \mathbf{V}}{\partial x}u + \frac{\partial \mathbf{V}}{\partial y}v + \frac{\partial \mathbf{V}}{\partial z}w + \frac{\partial \mathbf{V}}{\partial t} \tag{1.2.0.5}$$

At this point is is useful to introduce the so-called *material derivative*, or *substantial derivative*

$$\frac{D\mathbf{V}}{Dt} = \mathbf{a}_p \tag{1.2.0.6}$$

$$= \frac{\partial \mathbf{V}}{\partial x}u + \frac{\partial \mathbf{V}}{\partial y}v + \frac{\partial \mathbf{V}}{\partial z}w + \frac{\partial \mathbf{V}}{\partial t} \tag{1.2.0.7}$$

Or, using the notation $(\mathbf{V} \cdot \nabla)\mathbf{V}$ for the convective terms

$$\mathbf{a}_p = \frac{D\mathbf{V}}{Dt} = (\mathbf{V} \cdot \nabla)\mathbf{V} + \frac{\partial \mathbf{V}}{\partial t} \tag{1.2.0.8}$$

The first term represents movement due to convection (the steady part of the flow) and the time-derivative term represents the local acceleration due to unsteadiness in the flow. Eq. (1.2.0.8) is the equation used to compute the acceleration of fluid particles in the Eulerian description.

1.3 Kinematics

Fig. 1 illustrates the four types of motion that are possible for a 2D fluid segment. Any time-evolution of a fluid segment will be a combination of these four types of motion.

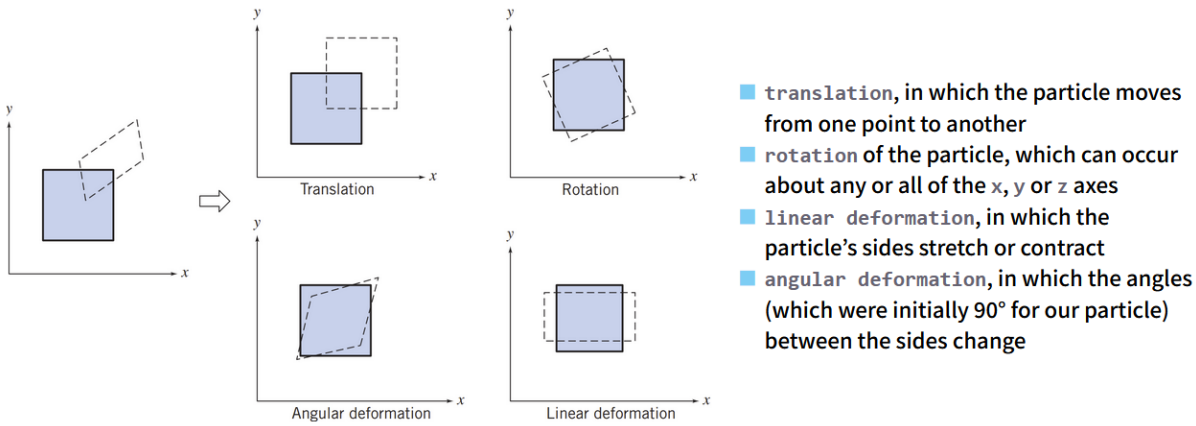

Figure 1: Various types of motion of a 2D fluid segment.

1.3.1 Translation

The translation of an individual fluid particle is connected with the velocity field via eq. (1.2.0.8)

$$\mathbf{a}_p = \frac{D\mathbf{V}}{Dt} = (\mathbf{V} \cdot \nabla)\mathbf{V} + \frac{\partial \mathbf{V}}{\partial t}$$

Since this can be solved to find the translation of an individual particle, it may be used to describe how fluid segments are translated as well. We see that translatory motion depends only on the acceleration of particles in the flow.

1.3.2 Rotation

The rotation of a particle in a flow can be modelled with a vector $\omega = (\omega_x, \omega_y, \omega_<)$ whose components are the angular velocities associated with rotation about the three coordinate axes. Fig. 2 a) and b) show how a square fluid segment undergoes some deformation and rotation during a time interval Δt. The deformation is described by the translation displacements $\Delta\eta$ and $\Delta\xi$ as well as the angular displacements $\Delta\alpha$ and $\Delta\beta$. Note that $\Delta\alpha$ is positive, while $\Delta\beta$ is negative, since we use the convention that counterclockwise rotation is the positive direction of rotation.

Fig. 2 shows

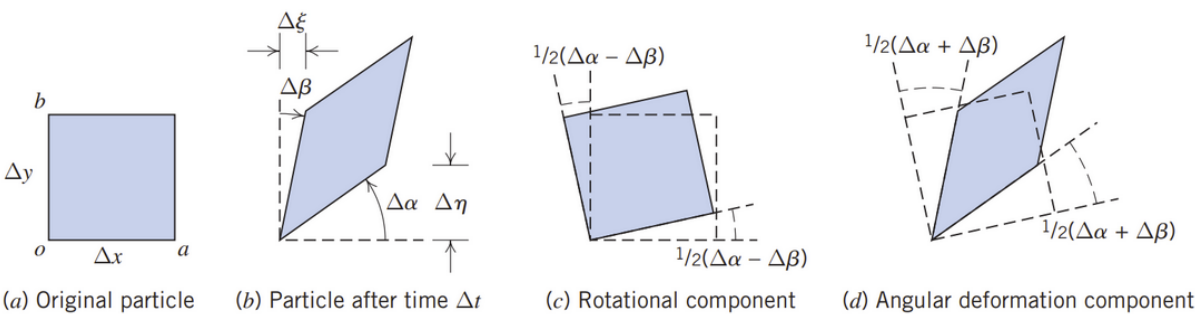

Figure 2

The segment's rigid body rotation (the rotational motion) can be found by taking the average of the two angular displacements, see fig. 2 c). The minus sign in $\frac{\Delta\alpha - \Delta\beta}{2}$ is

due to the fact that we are taking the average in the counterclockwise direction, and the counterclockwise rotation of ob is $-\Delta\beta$.

We will also need to find the translational displacements $\Delta\eta$ and $\Delta\xi$. To do so, we will consider the situation where all of the displacements and angles involved in the problem are small. Then we can Taylor expand the velocity field $\mathbf{V}$ to get

$$V_x(b) = V_x(o) + \frac{\partial V_x}{\partial y}\Delta y \tag{1.3.2.1}$$

So the x-displacement of b will be $\left(V_x(o) + \frac{\partial V_x}{\partial y}\Delta y\right)\Delta t$. To find $\Delta\xi$ we must subtract the displacement of the origin which is $V_x(o)\Delta t$. That leaves us with

$$\Delta\xi = \frac{\partial V_x}{\partial y}\Delta y \Delta t \tag{1.3.2.2}$$

The same argument can be used to obtain

$$\Delta\eta = \frac{\partial V_y}{\partial x}\Delta x \Delta t \tag{1.3.2.3}$$

In our small-displacement approximation, we can write the angular displacements as

$$\Delta\alpha = \frac{\Delta\eta}{\Delta x} \tag{1.3.2.4}$$

$$\Delta\beta = \frac{\Delta\xi}{\Delta y} \tag{1.3.2.5}$$

Leading to the following result:

$$\frac{1}{2}\frac{(\Delta\alpha - \Delta\beta)}{\Delta t} = \frac{1}{2}\frac{\left(\frac{\Delta\eta}{\Delta x} - \frac{\Delta\xi}{\Delta y}\right)}{\Delta t} = \frac{1}{2}\frac{\left(\frac{\partial V_y}{\partial x}\Delta t - \frac{\partial V_x}{\partial y}\Delta t\right)}{\Delta t} \tag{1.3.2.6}$$

To find the angular velocity about the z-axis, we take the limit $\Delta t \to 0$

$$\omega_z = \frac{1}{2}\left(\frac{\partial V_y}{\partial x} - \frac{\partial V_x}{\partial y}\right) \tag{1.3.2.7}$$

This is the z-component of the curl of the velocity field (multiplied by a factor of 1/2). The same argument can be applied to fluid segments in the yz- and xz-planes, so we conclude that

$$\omega = \frac{1}{2}\nabla \times \mathbf{V} = \zeta \tag{1.3.2.8}$$

Where we've introduced the vorticity $\zeta = 2\omega = \nabla \times \mathbf{V}$ to get rid of the factor of 1/2.

1.3.3 Angular Deformation

Fig. 1 d) illustrates how the angular deformation component is the sum of the individual angular displacements $\Delta\alpha + \Delta\beta$ (find out where the factor of 1/2 comes from!)

We can repeat the last steps of our previous calculation to see that the total angular deformation in the xy-plane E_{xy} is given by

$$E_{xy} = \lim_{\Delta t \to 0} \frac{1}{2} \frac{(\Delta\alpha + \Delta\beta)}{\Delta t} \tag{1.3.3.1}$$

$$= \frac{1}{2}\left(\frac{\partial V_y}{\partial x} + \frac{\partial V_x}{\partial y}\right) \tag{1.3.3.2}$$

And by the same reasoning

$$E_{yz} = \frac{1}{2}\left(\frac{\partial V_z}{\partial y} + \frac{\partial V_y}{\partial z}\right) \tag{1.3.3.3}$$

$$E_{xz} = \frac{1}{2}\left(\frac{\partial V_z}{\partial x} + \frac{\partial V_x}{\partial z}\right) \tag{1.3.3.4}$$

1.3.4 Linear Deformation

Linear deformation (deformation that preserves right angles) in the x_i-direction is only possible if $\frac{\partial V_i}{x_i} \neq 0$. This quantity represents the longitudinal strain rates in the x_i-direction. Unlike other types of deformation, linear deformation may lead to changes in volume given by the volume dilation rate

$$\text{volume dilation rate} = \frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z} = \nabla \cdot \mathbf{V} \tag{1.3.4.1}$$

A flow is *incompressible* if the volume dilation is zero $\nabla \cdot \mathbf{V} = 0$.

1.3.5 Tensor Representation of Deformation

To model the deformation that fluid segments undergo, we need to consider the gradient of the velocity field (we have just seen that the derivatives of the velocity field are needed to describe the various types of fluid segment motion).

The gradient of the velocity field is a tensor with the following entries

$$\nabla \mathbf{V} = A_{ij} = \partial_i v_j \tag{1.3.5.1}$$

It is a known result from tensor calculus that *any* tensor can be decomposed into a symmetric and an antisymmetric part. This is easily shown by writing the tensor T_{ij} in a clever way and adding and subtracting the transpose T_{ji}

$$\begin{aligned} T_{ij} &= \frac{1}{2}T_{ij} + \frac{1}{2}T_{ij} \\ &= \frac{1}{2}T_{ij} + \frac{1}{2}T_{ji} + \frac{1}{2}T_{ij} - \frac{1}{2}T_{ji} \\ &= \frac{1}{2}(T_{ij} + T_{ji}) + \frac{1}{2}(T_{ij} - T_{ji}) \end{aligned}$$